

HAT-P-36b

R-0829

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250219 · created 2026-07-28T16:52:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460725.9127 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.032
Transit depth (Rp/Rs)^2	1.5 +/- 0.79 [%]
Orbital Inclination (inc)	84.6 +/- 2.45
Airmass coefficient 1 (a1)	1.393 +/- 0.014
Airmass coefficient 2 (a2)	-0.31 +/- 0.15
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	5.88
Transit Duration (day)	0.0894 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.032 vs 0.126 (deviation 0.13σ)
Duration fit vs published	0.0894 d vs 0.0925 d (deviation 0.43σ)
Mid-time O–C	9.1 min
Depth SNR	3.8
Residual scatter	1.01 %

Light curve

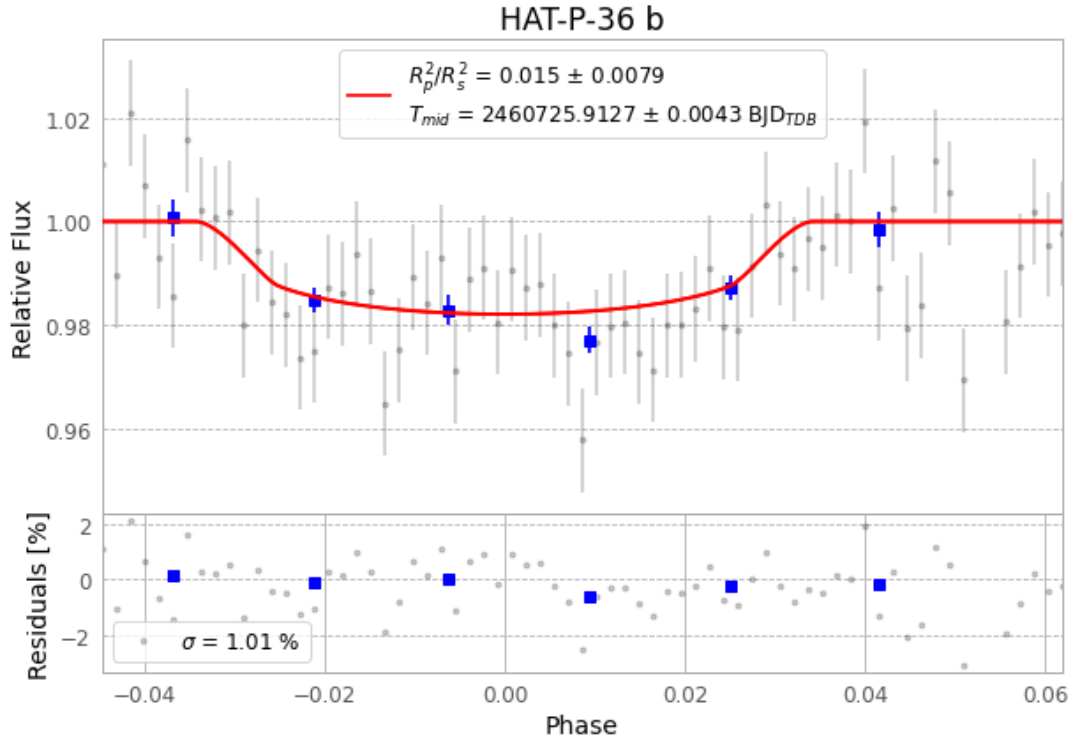


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-02-19.png (SHA-256 22b21bc1f0f7609e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f8ef08380189ec23...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

69 FITS frames (45 MB), HATP-36250219082115.FITS → HATP-36250219114515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250219.log (9e6c0c47862e...), FinalLightCurve_HAT-P-36 b_2025-02-19.png (22b21bc1f0f7...), FinalLightCurve_HAT-P-36 b_2025-02-19.csv (dc4609e73019...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 16:52Z.